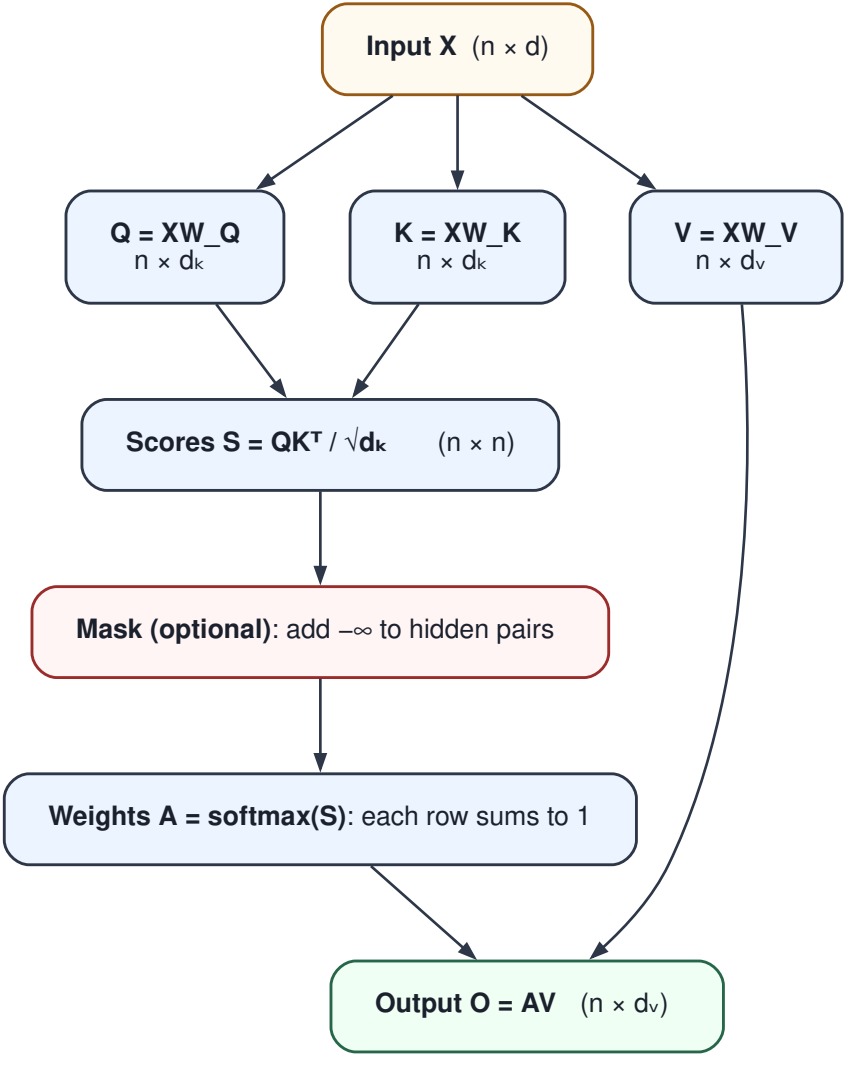
Input X (n × d)
Q = XW_Q
n × dk
K = XW_K
n × dk
V = XW_V
n × dv
Scores S = QKT / √dk (n × n)
Mask (optional): add −∞ to hidden pairs
Weights A = softmax(S): each row sums to 1
Output O = AV (n × dv)